

An LLM-based Waterway Scheduling Agent Integrated with Semantic Perception and Evolutionary Optimization

Junjie Weng¹, Mozi Chen¹, Yiling Ren¹, Yang Liu², Changzhen Li³, and Kezhong Liu¹

¹School of Navigation, Wuhan University of Technology, Wuhan, China

Email: {wengpfr, chenmz, renyl, kzliu}@whut.edu.cn

²Shenzhen Maritime Safety Administration, Shenzhen, China

Email: hillvictor@qq.com

³School of Information Engineering, Wuhan University of Technology, Wuhan, China

Email: changzhen.li@whut.edu.cn

Abstract—The exponential growth of global vessel traffic in restricted waterways poses severe challenges to navigational safety and scheduling efficiency. Traditional manual scheduling suffers from high latency under dynamic pressure, while existing heuristic algorithms often lack the situational awareness required to interpret unstructured Automatic Identification System (AIS) data. To address these challenges, this paper proposes an LLM-driven Waterway Scheduling Agent Framework. The core of this framework is a “numeric-to-semantic” interface that integrates dynamic situational awareness with an underlying Adaptive Genetic Algorithm (AGA). The system features a perception layer that translates physical kinematic deviations into human-readable semantic reports, enabling the LLM to act as a cognitive decision core. Rather than modifying physical coordinates directly, the LLM translates dynamic environmental conditions into adaptive penalty weights for the AGA’s objective function. To manage inherent LLM uncertainties in this safety-critical domain, the framework incorporates a deterministic conflict detection operator as a strict physical guardrail, facilitating a reliable, human-supervised (“human-on-the-loop”) collaborative paradigm. Experimental results demonstrate that, compared to manual scheduling, the framework reduces the Total Schedule Adjustment Time from 31.0 hours to 9.5 hours (a 69.4% improvement), validating its capability to transition waterway management from static planning to a reliable, situation-driven adaptive response.

Index Terms—Waterway Scheduling, Large Language Model (LLM), Adaptive Genetic Algorithm, Semantic Perception, Intelligent Transportation Systems

I. INTRODUCTION

Maritime transport underpins the global economy, yet the exponential growth in vessel traffic has turned core navigational channels into critical bottlenecks. To address these challenges, Agussurja et al. [1] modeled maritime traffic management as a Resource-Constrained Project Scheduling Problem (RCPSP), emphasizing that static planning often fails to address the dynamic contention for spatial resources. Furthermore, in the specific context of restricted waterways necessitating one-way traffic controls, Gan et al. [2] demonstrated that efficient online scheduling is indispensable for ensuring both navigational safety and channel capacity.

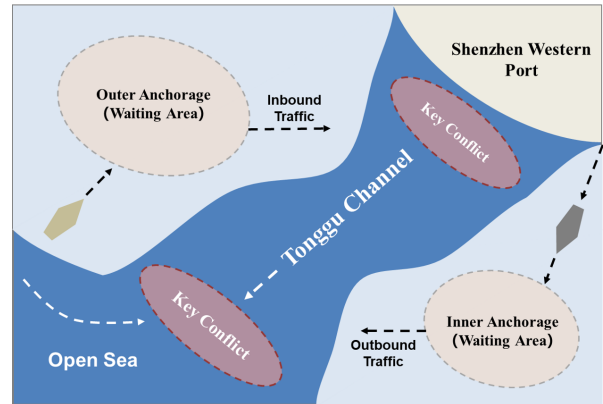


Fig. 1. Schematic diagram of the Shenzhen Tonggu Channel navigation environment. The diagram illustrates the critical “Key Conflict” zones where inbound and outbound traffic flows intersect with anchorage operations.

These operational constraints are acutely manifested in the Shenzhen Tonggu Channel. As illustrated in Fig. 1, this channel serves as the sole deep-water passage connecting the Shenzhen Western Port area to the open sea. The areas where inbound and outbound traffic intersect with anchorage operations form complex spatial bottlenecks. The channel operates under severe restrictions characterized by significant tidal influence, narrow navigation windows, and a high saturation of 200,000-ton ultra-large vessels. Currently, vessel scheduling in this high-stress environment relies primarily on the *expert heuristics* of senior dispatchers based on Vessel Traffic Service (VTS) declaration data. While effective for low-traffic scenarios, this manual paradigm exhibits critical limitations under high-density pressure: response latency is high, and planning redundancy is suboptimal, making it difficult to adapt to frequent, dynamic changes in the operational environment.

Existing automated research, primarily focused on mathematical programming and metaheuristic algorithms, attempts to mitigate these issues but faces significant hurdles in real-

world application. First, traditional algorithms lack *situational awareness*; they struggle to process unstructured streaming data generated by AIS, failing to capture the *operational semantics* behind anomalies like speed drops or trajectory deviations. Second, algorithm configurations are typically static—parameters such as crossover probabilities in genetic algorithms are predefined, rendering them incapable of adaptive tuning in response to real-time congestion risks. Finally, existing systems often operate as “black-box” solvers, lacking the natural language interface required for transparent, human-supervised decision-making. Rather than requiring dispatchers to manually adjust algorithmic parameters or manually re-sequence vessels, our system functions as a cognitive assistant. It generates human-readable semantic reports and interpretable scheduling schemes, effectively keeping the human operator “on the loop” for final review, verification, and approval, thereby establishing a practical collaborative paradigm.

To address these challenges, this paper proposes an LLM-driven Waterway Scheduling Agent Framework. The core technical contribution of this work lies in establishing a “numeric-to-semantic” interface that integrates dynamic situational awareness with evolutionary optimization. Rather than treating dynamic scheduling as a purely mathematical constraint-solving problem, we frame it as a closed-loop perception–reasoning–execution system. Specifically, the core contributions are as follows:

- 1) **AIS Semantic Parsing Layer** : We design a perception module that translates physical kinematic deviations (e.g., ETA drift, speed fluctuations) into human-readable semantic situational reports. This mechanism bridges the gap between raw AIS data streams and high-level scheduling logic, enhancing both situational awareness and operational interpretability.
- 2) **LLM-Driven Cognitive Decision Core**: We propose an agentic architecture where a domain-adaptive LLM acts as the cognitive center. By evaluating semantic reports against a structured domain knowledge base, the LLM interprets anomalies and orchestrates the underlying optimization tools, transitioning the scheduling paradigm from static empirical planning to a situation-driven adaptive response.
- 3) **Context-Aware Parameter Adaptation for Evolutionary Execution**: We introduce a mechanism to couple natural language reasoning with mathematical execution. Rather than using static configurations, the LLM evaluates dynamic environmental conditions and translates them into adaptive penalty weights for the underlying Genetic Algorithm (GA). This allows high-level operational priorities to be reflected in the algorithmic search space, facilitating a better balance between safety and efficiency under traffic pressure.

II. RELATED WORKS

A. System Scheduling Modeling and Heuristic Algorithms

Waterway scheduling is inherently a complex optimization problem constrained by multiple spatiotemporal factors. Early

research primarily focused on establishing precise mathematical models, such as the mathematical programming model proposed by Li and Jia [3]. For seaport scheduling under tidal effects, Wang et al. [4] developed tidal phase coordination models that integrate dynamic tidal window constraints into vessel scheduling optimization. However, when addressing high-dimensional real-time constraints, exact algorithms often encounter the “curse of dimensionality,” rendering them computationally infeasible for large-scale scheduling problems. Consequently, meta-heuristic algorithms have been widely adopted due to their ability to provide near-optimal solutions with reasonable computational effort [5]. Al Samrout et al. [6] developed an improved GA, to solve dynamic berth scheduling with integrated transshipment operations, showing significant computational efficiency and solution quality improvements compared with traditional approaches, and Cao et al. [7] employed emerging algorithms, including the cuckoo search, to enhance both solution quality and convergence efficiency. Although these approaches excel in static scenarios, their inherent ‘black-box’ nature hinders the system’s ability to comprehend the operational semantics behind AIS trajectories, resulting in a lack of immediate cognitive response to real-time dynamic disturbances.

B. LLM-based Autonomous Agents in Maritime Contexts

Autonomous Agents centered on LLMs have exhibited exceptional potential for perception and decision-making in complex control tasks.

In terms of professional knowledge retention and decision legitimacy, Pei et al. [8] evaluated the mastery of ship maneuvering theory and skills by LLMs. They demonstrated that the model is capable of integrating abstract navigation theories with specific operational scenarios, achieving a leap from mere text retrieval to deep semantic reasoning.

Regarding real-time situational awareness and the semantic transformation of AIS data, the SEMINT framework [9] presented an LLM-enhanced method for long-term vessel trajectory prediction. This model performs semantic modeling on heterogeneous AIS data, transcribing coordinate sequences into human-comprehensible navigation scenarios. Similarly, Guo et al. [10] proposed a natural language-based AIS trajectory query method, realizing a cross-dimensional mapping from physical coordinates to vessel behavioral semantics. These studies lay the technical groundwork for identifying waterway anomalies and translating them into prompts comprehensible to LLMs in this work.

Furthermore, Sun et al. [11] proposed the VTS-LLM framework. By introducing domain-adaptive LLM agents and incorporating the autonomous invocation of external professional tools, they employed the LLM as a cognitive core to enhance situational awareness and drive closed-loop operations.

In the intelligent agent framework constructed in this paper, the LLM functions as the cognitive core. By integrating real-time AIS semantics via Prompt Engineering and leveraging Tool-use to orchestrate the underlying Genetic Algorithm

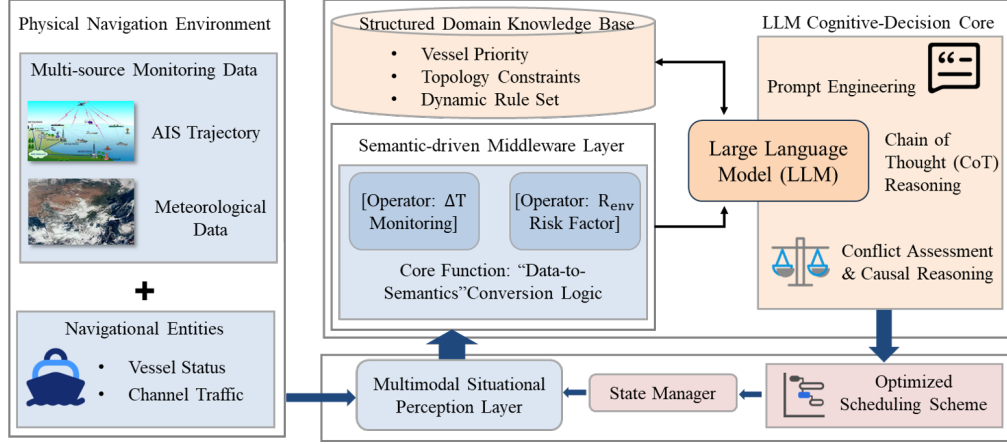


Fig. 2. The overall architecture of the closed-loop intelligent agent system for waterway scheduling, integrating four collaborative layers driven by an LLM cognitive core.

(GA), the system achieves closed-loop rescheduling. This collaborative “Perception-Reasoning-Execution” architecture establishes a novel paradigm for addressing the flexibility challenges inherent in waterway scheduling within dynamic environments.

III. METHODOLOGY

A. Overall System Architecture

This study constructs a closed-loop waterway scheduling agent system designed to achieve full-process automation [12], ranging from multi-source environmental monitoring to the automatic reconstruction of scheduling plans. As illustrated in Fig. 2, this architecture transcends the limitations of traditional scheduling systems, which are typically confined to processing structured numerical data. By incorporating the LLM as the cognitive core, the system is deeply integrated into four collaborative layers:

- 1) **Multi-modal Situational Perception Layer:** This layer dynamically integrates real-time vessel AIS trajectory data and environmental monitoring data via dedicated interfaces.
- 2) **Semantic-driven Middleware:** Serving as the core for transforming physical data into logical decisions, this layer functions beyond merely monitoring data deviations. It executes “data-to-semantic” translation logic, mapping numerical perturbations into natural language situational reports enriched with domain-specific context, thereby preparing comprehensible decision inputs for the LLM.
- 3) **LLM Cognitive-Decision Core:** Built upon the Dify Agent framework, this layer integrates domain-adaptive LLMs with expert knowledge bases. Through Prompt Engineering, it performs conflict assessment and causal reasoning on semantic reports. It autonomously triggers specific scheduling intervention tools while strictly adhering to constraints regarding one-way navigation and meteorological safety [13].

- 4) **Adaptive Optimization Execution Layer:** Based on the structured instructions generated by the decision core, this layer invokes a Genetic Algorithm engine equipped with adaptive penalty functions to perform large-scale parallel optimization [14]. The execution results are then fed back via a State Manager to achieve a closed-loop system.

The pivotal advantage of this closed-loop design lies in its ability to transmit every anomalous fluctuation in the waterway environment and navigational entities to the cognitive layer in real-time via a Semantic Stream. This capability drives the system’s evolution from “static empirical scheduling” to a “situation-driven adaptive response,” thereby significantly enhancing the allocation efficiency of waterway resources in complex, dynamic environments.

B. Semantic Mapping Based on Multi-modal Situational Perception

To achieve the decoupling of low-level physical situations from high-level cognitive decision-making, this section designs a Multi-source Situational Perception and Semantic Mapping Mechanism. By establishing monitoring operators for AIS kinematic deviations and meteorological environmental fluctuations, this mechanism transforms numerical fluctuations into semantic representations comprehensible to the LLM.

AIS Kinematic Deviation Monitoring System. By utilizing real-time AIS trajectory sequences, we extract current vessel motion characteristics and compare them against the original declaration plan. The specific definitions and sample values of the AIS input state vectors used in this perception layer are detailed in Table I. We define the time deviation operator ΔT to capture anomalies in vessel inbound progress [15]:

$$\Delta T = |ETA_{\text{real}} - ETA_{\text{plan}}| \quad (1)$$

where ETA_{real} represents the Estimated Time of Arrival (ETA) projected via the real-time average speed V_{avg} and the remaining voyage distance D_{remain} :

$$ETA_{\text{real}} = T_{\text{now}} + \frac{D_{\text{remain}}}{V_{\text{avg}}} \quad (2)$$

When $\Delta T > \theta_{\text{ais}}$ (where θ_{ais} denotes a preset time-trigger threshold), the Semantic Middleware Layer generates a kinematic anomaly report indicating the risk of vessel delay.

TABLE I
SPECIFICATION OF AIS-BASED INPUT STATE VECTORS FOR THE
SITUATIONAL PERCEPTION LAYER

Attribute	Semantic Definition	Sample Value
<i>_id</i>	Unique identifier for the DB record	1876660184379969537
<i>shoreMmsi</i>	MMSI of the shore-based station	4130744
<i>mmsi</i>	Maritime Mobile Service Identity	413400020
<i>receiveTime</i>	Precise timestamp of acquisition	1736265600052
<i>latitude</i>	Geodetic latitude of the vessel	30.1565
<i>longitude</i>	Geodetic longitude of the vessel	122.1428
<i>rot</i>	Rate of Turn (ROT) [deg/min]	2.3
<i>sog</i>	Speed Over Ground (SOG) [knots]	1.0
<i>trueHeading</i>	True Heading of the hull [deg]	268
<i>cog</i>	Course Over Ground (COG) [deg]	29

Meteorological Environmental Situational Perception.

Unlike traditional scheduling approaches that focus exclusively on vessel dynamics, this system incorporates a meteorological situational perception module to address the impact of complex weather conditions on navigational capabilities. The system monitors wind speed W_s and visibility V_v in real-time, defining the Environmental Risk Factor R_{env} as follows:

$$R_{\text{env}} = \begin{cases} \text{High Risk,} & \text{if } W_s > W_{\text{limit}} \text{ or } V_v < V_{\text{limit}} \\ \text{Normal,} & \text{otherwise} \end{cases} \quad (3)$$

where W_{limit} represents the maximum safe wind speed for large vessels, and V_{limit} denotes the visibility threshold for channel closure.

Multi-source Semantic Aggregation and Report Generation. The Semantic-driven Middleware performs spatiotemporal alignment and correlation analysis on the aforementioned discrete physical indicators [16]. It then encapsulates them into natural language templates and transmits them to the LLM Cognitive Core. The aggregation logic follows the equation [17]:

$$\text{Report}_{\text{semantic}} = \Phi(\text{Ship}_{\text{info}}, \Delta T, R_{\text{env}}) \quad (4)$$

Semantic Mapping Example:

- Input: $W_s = 12\text{m/s}$, Ship Name = “MSC ZOE”, $\Delta T = 85\text{min}$.
- Output Report: “Current navigation environment is High Risk (Wind Speed Exceeded). Vessel ‘MSC ZOE’ has significantly decelerated due to strong winds, with a delay of 85 minutes, expected to trigger time-window overlap conflicts at the channel exit.”

Through this mechanism, the system achieves a transition from “numerical monitoring” to “semantic understanding,” providing logical facts enriched with operational context for subsequent knowledge-based reasoning.

C. Domain-Adaptive LLM Reasoning Engine with Multi-modal Perception

This section elaborates on the decision-making mechanism of the agent. The inference engine is constructed upon a LLM. By integrating and aligning multi-modal semantic reports with a structured domain knowledge base, it achieves a logical transition from “situational awareness” to “scheduling decision-making,” thereby generating professional and interpretable decisions.

Domain Knowledge Base and RAG. To ensure the compliance of scheduling schemes, the system integrates three categories of core knowledge bases and transforms them into cognitive constraints for the LLM [18]:

- 1) Classification of vessels and Priority Configuration: This defines a priority weighting system ranging from combat-readiness vessels (Level 1) to general cargo ships (Level 7). Based on this, the LLM determines high-value voyages that must be prioritized under resource-constrained conditions.
- 2) Spatial Topology and Berth Constraints: This provides mapping relationships for port berths, enabling the LLM to comprehend the geographic distribution of different operational zones and avoid cross-zone conflicts.
- 3) Dynamic Scheduling Rule Set: Serving as the core criterion for decision-making, this set includes seasonal definitions of day and night, standard safety intervals, tugboat operation windows, and various penalty and reward mechanisms.

Decision Reasoning Driven by Semantic Reports. Leveraging the *Chain of Thought* (CoT) technique, the inference engine orchestrates a reasoning process upon receiving multimodal semantic reports containing meteorological and AIS anomalies [19].

It first executes Situational Interpretation and Contextual Rule Activation. Specifically, when the semantic monitoring module identifies environmental risks or temporal deviations, the system automatically retrieves and activates the corresponding domain constraints.

Subsequently, the process advances to the multi-criterion Decision and Arbitration phase. Here, the agent conducts an in-depth analysis of the cascading effects of delayed vessels on the subsequent inbound sequencing, performing a comprehensive evaluation and deduction by synthesizing vessel priority weights.

Algorithm 1: LLM-Constrained Adaptive Genetic Scheduling

Input: Real-time State S_t , Semantic Report R_{sem} (from LLM)
Output: Optimized Schedule X^*

```

/* Step 1: Parse Semantic Constraints */
1 Extract Constraint Set  $C_{sem} \leftarrow \text{Parse}(R_{sem})$ 
2 Identify High-Priority Vessel set  $V_{prio}$  based on  $C_{sem}$ 
3 Initialize Penalty Coefficient  $\alpha_p$  and  $G_{max}$ 
/* Step 2: Initialize Population */
4  $P_0 \leftarrow \text{InitializePopulation}(S_t, \text{Encoding}=\text{Direct})$ 
5  $g \leftarrow 0$ 
6 while  $g < G_{max}$  do
/* Step 3: Evolution Loop */
7   foreach individual  $X_i \in P_g$  do
8     Calculate Soft Score  $Score_{soft}(X_i)$ 
9     Calculate Hard Violation  $Penalty_{hard}(X_i)$ 
/* Adaptive Weight Calculation (Eq. 6) */
10     $w_{adaptive} \leftarrow (g/G_{max}) \cdot \alpha_p$ 
/* LLM-Injected Priority Logic */
11    if  $X_i$  violates  $V_{prio}$  constraints then
12       $Penalty_{hard}(X_i) \leftarrow Penalty_{hard}(X_i) \times 1.5$ 
13    end
/* Final Fitness Calculation (Eq. 5) */
14     $F(X_i) \leftarrow Score_{soft}(X_i) - w_{adaptive} \cdot Penalty_{hard}(X_i)$ 
15  end
16   $P_{parents} \leftarrow \text{Selection}(P_g, F)$ 
17   $P_{offspring} \leftarrow \text{CrossoverAndMutation}(P_{parents})$ 
18   $P_{g+1} \leftarrow P_{offspring}$ 
19   $g \leftarrow g + 1$ 
20 end
21 return Best individual  $X^*$  from  $P_{G_{max}}$ 

```

Based on the outcomes of logical evolution, the inference engine adaptively matches the optimal intervention mode from the agent's tool repository [20]: For isolated single-dimensional spatiotemporal deviations, it activates the Local Fine-tuning mode to execute precise timestamp calibration for specific vessels; conversely, in the face of systemic disruptions induced by sudden meteorological shifts, it transitions to the Global Restructuring mode. This triggers the Adaptive Genetic Algorithm to achieve global temporal sequencing optimization and schedule reconstruction under complex constraints.

D. Situation-Driven Evolutionary Optimization and Schedule Reconstruction

Upon receiving scheduling intervention directives from the Cognitive Decision Core, the system transitions into the execution phase. This section elucidates the mechanism for translating high-level decision logic generated by the LLM into evolutionary constraints for the underlying algorithm and subsequently realizing the dynamic reconstruction of scheduling schemes via an improved Adaptive Genetic Algorithm (AGA). The complete pseudocode of this LLM-constrained optimization process is presented in Algorithm 1, which details how semantic constraints are parsed and injected into the evolutionary loop.

The system eschews direct responses to ambiguous textual instructions, instead employing a Parameter Parsing Engine to map the analytical conclusions of the decision core into explicit constraints for the underlying evolutionary algorithm. This approach achieves a deep coupling between semantic logic and physical execution. This mapping process is driven by two core dimensions: First is Environmental Constraint Dynamics, which automatically activates corresponding navigation rule operators based on semantic results derived from multi-modal perception. Second is the Disturbance-Driven Reconstruction Strategy. Upon detecting a significant drift in the ETA, the system—while synchronizing physical states—outputs differentiated reconstruction directives based on the magnitude of the deviation.

For minor delays inducing minimal conflict, the system executes time-axis shifting and local reordering, achieving rapid alignment by correcting specific timestamps. Conversely, for systemic conflicts resulting in severe berth or channel window overlaps, it initiates conflict resolution and global iteration. This involves extracting priority weights W_i from the vessel definition library and injecting them into the penalty term of the evolutionary algorithm's objective function, thereby ensuring the scheduling security of high-value voyages under resource competition.

Following the mainstream algorithmic architecture in maritime scheduling, this study adopts an improved direct-encoding Adaptive Genetic Algorithm (AGA). While various optimization methods, ranging from exact mathematical programming to other meta-heuristics, have been proposed for scheduling tasks, the selection of a GA in this framework is primarily driven by its structural compatibility with the semantic module.

Specifically, restricted waterway scheduling involves strict discrete time-window constraints. The discrete encoding strategy of a GA accommodates these constraints efficiently, helping to mitigate the computational overhead often encountered by exact mathematical solvers when handling complex, high-density traffic scenarios. Furthermore, its penalty-based constraint handling mechanism provides a straightforward pathway to incorporate the semantic outputs of the LLM. Instead of requiring complex mathematical reformulations for every operational rule change, the high-level semantic priorities parsed by the LLM (e.g., temporary vessel priority adjustments) can be translated directly into numerical weights within the objective function's penalty terms. This characteristic facilitates the integration of natural language reasoning with mathematical execution.

Based on this rationale, the scheduling scheme is encoded as a real-number vector $X = (x_1, x_2, \dots, x_N)$, where x_i represents the index of the position selected by vessel s_i within its discretized time window candidate list. This encoding strategy transforms the continuous-time optimization problem into a discrete combinatorial optimization problem, effectively compressing the solution space and ensuring the operational efficacy of genetic operators.

The fitness function is pivotal in guiding the algorithm's

search process. One of the core innovations of this paper is the design of an adaptive penalty fitness function, defined as follows:

$$F(S) = \text{Score}_{\text{soft}}(S) - w_{\text{adaptive}}(g) \cdot \text{Penalty}_{\text{hard}}(S) \quad (5)$$

Here, $\text{Score}_{\text{soft}}(S)$ denotes the soft constraint score term, used to evaluate the economy and efficiency of the scheduling scheme; $\text{Penalty}_{\text{hard}}(S)$ represents the hard constraint penalty term, quantifying the degree of violation regarding core constraints such as safety rules. The crux of this function lies in the adaptive weight $w_{\text{adaptive}}(g)$, which increases dynamically with the evolutionary generation g :

$$w_{\text{adaptive}}(g) = \frac{g}{G_{\text{max}}} \cdot \alpha_p \quad (6)$$

where G_{max} is the maximum number of evolutionary generations, and α_p is the penalty growth coefficient.

By imposing a smaller penalty weight in the early stages of evolution, this mechanism allows the algorithm to conduct extensive exploration within a solution space that may contain partial mild constraint conflicts, effectively preventing premature convergence. As iterations proceed and the weight increases linearly, the algorithm gradually intensifies the elimination pressure on infeasible solutions. This drives the population to converge towards optimal regions within the feasible domain, thereby achieving fine-grained exploitation in the later stages.

The global optimization process driven by the Genetic Algorithm achieves system reconstruction through a rigorous closed-loop logic. Upon acquiring the optimal individual, the Conflict Detection Operator immediately intervenes to execute a secondary verification. This step conducts a final safety audit and topological consistency check on the generated scheme, ensuring the integrity and physical feasibility of the scheduling logic. Facilitated by the State Manager, which handles state persistence and visual transformation, the system updates the scheduling baseline in real-time and generates fine-grained timeline adjustment reports.

Finally, a semantic summary of the reconstructed plan is fed back to the Cognitive Layer Review Mechanism. Here, the decision agent assesses the alignment between the optimization results and the initial semantic demands of the perception layer, thereby establishing a comprehensive closed-loop pipeline spanning “Situational Representation,” “Evolutionary Decision,” and “Cognitive Calibration.”

E. Uncertainty Management and System Guardrails

While Large Language Models demonstrate strong cognitive reasoning capabilities, their deployment in safety-critical domains such as waterway scheduling necessitates rigorous uncertainty management to mitigate inherent risks like hallucination or semantic misinterpretation. To ensure absolute operational safety, our agent framework strictly adheres to a “Propose-and-Verify” paradigm, establishing deterministic guardrails at the execution layer.

The LLM is explicitly restricted from directly modifying physical vessel coordinates or timestamps. Instead, it operates exclusively in the semantic and parameter space, outputting scheduling intents and adaptive penalty weights (w_{adaptive}). This architectural isolation fundamentally shields the physical kinematic execution from the stochastic nature of the LLM.

At the execution level, a deterministic Conflict Detection Operator acts as the ultimate physical guardrail. Upon the completion of the GA optimization driven by the LLM’s parameters, this operator conducts a rigid spatiotemporal topology audit. It mathematically verifies whether the generated schedule violates any hard constraints, such as one-way navigation rules, minimum safety intervals, or tidal window boundaries. If the LLM’s initial parameter adjustments (derived from hallucinated or misparsed semantic reports) lead to an infeasible region, the detection operator will intercept the output.

To address these intercepted outputs or severe uncertainties, the framework incorporates an automated fallback mechanism. If the LLM generates unparsable instructions, or if the resulting schedule fails the deterministic conflict audit, the system automatically rejects the LLM’s proposal. In such boundary cases, the agent triggers a conservative “First-Come-First-Served (FCFS)” baseline or immediately alerts human dispatchers to intervene. This fallback strategy ensures that the waterway safety baseline is never compromised by algorithmic errors or cognitive uncertainties.

IV. EXPERIMENTS

To validate the perception and reconstruction performance of the closed-loop intelligent scheduling agent under dynamic and uncertain conditions, this study establishes a high-fidelity simulation environment grounded in real-world waterway data. Furthermore, we design stress testing experiments incorporating multiple perturbations.

A. Experimental Environment and High-Stress Scenario Construction

This experiment selects the Shenzhen Tonggu Channel as the physical mapping object [21]. As a critical deep-water passage in the Pearl River Estuary, this channel exhibits typical dual spatiotemporal constraints: Spatially, it is restricted by channel width, necessitating strict one-way traffic control; temporally, the transit windows for large vessels are highly dependent on tidal levels and must adhere to safety intervals that differentiate between day and night. This highly constrained environment provides an ideal testbed for evaluating the agent’s boundary decision-making capabilities.

The simulation spans 24 hours, with the timeline discretized into 48 time slots of 30 minutes each. System parameter configuration follows standard settings for the Adaptive Genetic Algorithm: Population Size is set to 80, Maximum Generations to 200, and a direct encoding mechanism based on time window indices is enabled to ensure search efficiency.

To systematically assess the agent’s robustness under extreme navigation pressure, this study does not rely solely

on normalized historical data. Instead, we constructed an Augmented Dataset based on VTS declaration records from a typical workday. Considering that adverse weather often leads to reduced navigation efficiency—triggering an equivalent “high-density congestion” effect—we implemented an adversarial data augmentation strategy on the original traffic flow [22]:

- 1) Structured Pressure Injection: We significantly increased the proportion of ultra-large vessels with stringent navigation requirements, raising their share in the scheduling queue to over 40%.
- 2) Constraint Space Compression: A high percentage of large vessels implies longer safety intervals and stricter tidal windows. This drastically compresses the feasible solution space, forcing the algorithm to seek conflict-free solutions within extremely narrow temporal gaps.

This high-stress scenario aims to simulate boundary conditions where the “Multi-modal Perception Module” identifies environmental deterioration. In this scenario, any minor scheduling error could trigger cascading rule violations. By benchmarking system performance under such limit conditions, we can objectively verify the superiority of the proposed “Cognitive-Driven and Adaptive Execution” architecture over traditional methods.

B. Dynamic Response Mechanism Based on Real-time AIS Semantic Perception

This experiment aims to validate the agent’s closed-loop decision-making capability driven by multi-modal semantics. We designed a typical coupled meteorological-traffic perturbation scenario to simulate the entire process where severe sea conditions propagate through physical links to the information system, triggering the agent’s adaptive reconstruction.

A sudden perturbation is introduced at simulation time $T = 08 : 00$. The target vessel, “Yan zhan 98” (a bulk carrier), experiences a speed consistently lower than the planned value during its inbound voyage due to external *force majeure*.

Physical Monitoring: The system’s AIS perception module tracks the vessel’s Speed Over Ground (SOG) and Course Over Ground (COG) in real-time. Monitoring data indicates a significant deviation between the vessel’s real-time motion state and its declared plan, resulting in a substantial drift in its ETA.

Semantic Extraction: When the cumulative deviation ΔT breaches a preset threshold, the semantic middleware ceases to output raw coordinate streams. Instead, it synthesizes the following structured semantic report and pushes it to the Cognitive Core:

[Alert]: Vessel “Yanzhan 98” detected significant kinematic anomaly. Real-time ETA delayed by 120 minutes. Status: Active Deviation.

Upon receiving the semantic report regarding the 2-hour delay of “Yanzhan 98,” the LLM Cognitive Core initiates a CoT process to assess the systemic risks triggered by this single perturbation:

- 1) Cascading Conflict Prediction: Analyzing the current scheduling status, the LLM discovers that if “Yanzhan 98” is postponed by 2 hours, it will directly encroach upon the time window of the subsequent inbound vessel, “MSC RIFAYA.”
- 2) Semantic-based Value Arbitration: The LLM retrieves the vessel attribute knowledge base for semantic comparison. It identifies “MSC RIFAYA” as a high-priority container ship, while “Yanzhan 98” is a low-priority bulk carrier.
- 3) Decision Generation: Based on the above reasoning, the agent determines that a simple “First-Come-First-Served (FCFS)” postponement strategy would compromise the high-value vessel. Consequently, the system generates a “Global Restructuring” directive, requiring the GA engine to enforce the priority right of way for high-weight vessels via penalty functions during rescheduling.

To provide a transparent view of this cognitive process, Table II details the step-by-step execution workflow for the “Yan Zhan 98” scenario. It explicitly maps the transformation from the initial semantic intent to the final tool chain orchestration, demonstrating how the agent autonomously invokes the state manager and GA optimizer to resolve the conflict. Upon receiving the reconstruction directive, the Adaptive Genetic Algorithm completed global optimization within 40 seconds. The generated reconstruction scheme successfully postponed the inbound time of “Yanzhan 98” while leveraging scheduling gaps to ensure the punctuality of “MSC RIFAYA” and subsequent high-priority vessels. Consequently, the system effectively transformed “deviation risks” at the AIS data layer into “controllable adjustments” at the operational layer.

When facing such a sudden 2-hour AIS delay, traditional manual scheduling often adopts mechanical postponement, resulting in a cumulative Total Schedule Adjustment Time as high as 31.0 hours. Benefiting from semantic-perception-driven global optimization, the reconstruction scheme generated by this system limited the Total Schedule Adjustment Time to 10.5 hours. Experiments demonstrate that even without human intervention, the agent—relying solely on the semantic mapping of AIS data—reduced the systemic violation costs caused by sudden perturbations by 66.1%, thereby validating its adaptive robustness under dynamic monitoring.

C. Global Conflict Resolution Based on Semantic Priority

This experiment aims to validate the agent’s global optimization capability in handling high-density heterogeneous traffic flow by parsing the social attribute semantics of static AIS information, thereby surpassing traditional FCFS rules.

The experiment selects a peak traffic period within the simulation timeline, during which the channel schedule is in a highly saturated state. At this juncture, the target vessel “Shen Kuo” applies to enter the Tonggu Channel. AIS broadcasts identify this vessel as a “Scientific Research Vessel,” and its planned navigation window conflicts spatiotemporally with multiple en-route merchant ships.

TABLE II
OPERATIONAL MECHANISM OF THE SEMANTIC-DRIVEN
COGNITIVE DECISION LOOP

Phase	Process Content & System Logic
1. User Cmd	[Natural Language Instruction] "Postpone the scheduled time of vessel 'Yan Zhan 98' by 2 hours."
2. Agent Reasoning	[Step 1: Semantic Parsing] • <i>Intent</i>: Time_Adjustment • <i>Extraction</i>: {Target: "Yan Zhan 98", Action: "Delay", Value: "2.0h"} [Step 2: Task Planning] 1. Update State → 2. Global Optimization → 3. Feasibility Check
3. Tool Chain Action	[Step 3: Tool Chain Execution] 1. state_manager.update_ship("Yan Zhan 98", 2.0) ↪ <i>Return</i>: State Updated. 2. ga_optimizer.run_optimizer(constraints=new_state) ↪ <i>Return</i>: Optimization Complete. 3. conflict_checker.check_conflicts(schedule=opt_res) ↪ <i>Return</i>: New Schedule Validated.
4. System Feedback	[Step 4: Response Generation] "Command Executed. Schedule Adjusted." • Status: Success • Metric: Total Adj. Time = 10.5h • Result: Priority violation resolved."

Upon reading the static AIS data via the physical perception layer, the semantic middleware combines it with the domain knowledge base to perform a value assessment, generating the following high-level semantic description:

[Info]: Vessel 'Shen Kuo' identified as Special Task Vessel. Semantic Priority: Level 2 (High). Operational constraint: High time-sensitivity.

The system identifies that although this vessel is not a combat-ready warship, it belongs to the category of high-value scientific research task vessels, and its navigational efficiency takes precedence over ordinary cargo vessels. Targeting this conflict scenario, we compared the distinct decision-making logic of the intelligent agent against traditional manual scheduling:

Baseline: Senior dispatchers or traditional FCFS algorithms follow the principle of local avoidance. Since the current channel time slots are fully occupied, to ensure absolute safety, the baseline scheme chooses to postpone the "Shen Kuo" until the idle period after the peak ends.

However, the agent-based approach, driven by cognitive reasoning with semantic priority, achieved a leapfrog improvement in scheduling quality (as shown in Fig. 3). The research vessel "Shen Kuo," which was delayed by 10 hours in the baseline plan to ensure safety, was prioritized and

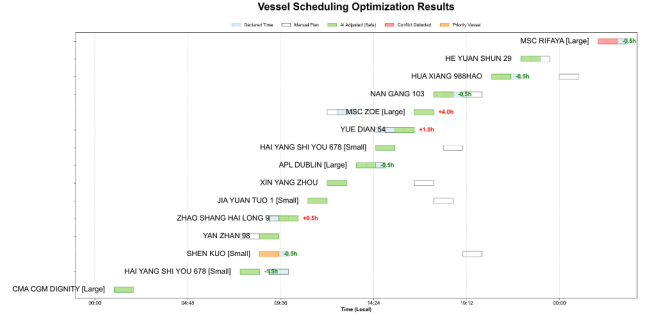


Fig. 3. Gantt chart of the optimized vessel scheduling scheme generated by the proposed Agent-based algorithm. The horizontal axis represents the local time, and the vertical axis lists the scheduled vessels. (1) Schedule Comparison: The light blue bars indicate the original declared time windows, while the hollow black boxes represent the baseline manual plan. The solid green bars ("AI Adjusted") denote the final optimized schedule, confirming that potential conflicts in the manual plan have been resolved into safe passage intervals. (2) Priority Handling: Orange bars highlight priority vessels (e.g., Shen Kuo) that are granted precedence in the sequence.

advanced by 1.5 hours in the optimized scheme. It was precisely embedded into a computationally verified structural idle window without disrupting the navigation of "Yan Zhan 98." Compared to the severe 10-hour delay in the baseline, the agent achieved a 1.5-hour advance through global window reconstruction, resulting in a net time advantage of 11.5 hours that effectively secured the mission timeliness of this high-value research vessel. This optimization was not limited to local adjustments; posterior analysis by the conflict detection module further verified the achievement of a Global Pareto Improvement. Without triggering rule violations or substantial delays for other en-route vessels, the system mined implicit gaps in the spatiotemporal structure to achieve Pareto-optimal allocation of navigation resources while ensuring safety. These experimental results confirm that the LLM-based agent possesses expert-like global vision, capable of precisely parsing AIS semantics to identify the "social value" of vessels, and ultimately realizing the global maximization of navigation efficiency through Coordinated Optimization.

D. Comprehensive Discussion and Computational Efficiency

The aforementioned micro-level case study confirms the agent's decision-making intelligence in handling single-point perturbations. This section further evaluates the architecture's computational efficiency and overall operational effectiveness from a system engineering perspective when handling large-scale concurrent tasks, thereby validating its deployment potential in real-world port VTS scenarios.

Experimental data iWei2022ChainOfThought indicates that when processing high-density scheduling tasks involving more than 15 vessels, the proposed agent incurs an average end-to-end algorithmic delay of less than 40 seconds—from receiving the AIS semantic stream to generating a conflict-free Gantt chart (as shown in Fig. 4). Given that ultra-large vessels possess immense physical inertia and typically require tens of minutes to execute meaningful speed or heading adjustments,

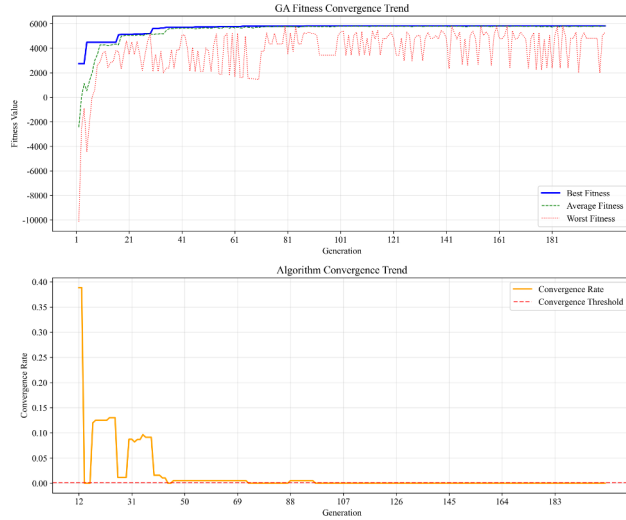


Fig. 4. Convergence analysis of the Genetic Algorithm. The upper plot illustrates the fitness evolution, where the best fitness (blue line) stabilizes rapidly after approximately 40 generations. The lower plot confirms computational efficiency, as the convergence rate (orange line) drops and remains below the threshold.

a system latency of under 40 seconds strictly satisfies the real-time responsiveness required for proactive traffic management. This ensures that VTS operators have ample time to review and broadcast the updated schedule before critical spatiotemporal conflict points are reached. In contrast, senior dispatchers often require tens of minutes for iterative reasoning and manual coordination when handling complex conflicts of the same scale, which frequently leads to missed intervention windows. This "second-level responsiveness" stems from the system's hierarchical architecture: the LLM cognitive center rapidly prunes the solution space, preventing the underlying AGA from traversing invalid regions. This significantly reduces convergence time while guaranteeing solution quality, demonstrating that the framework possesses the engineering capability to meet the VTS center's requirements for Real-time Dynamic Re-planning in response to unexpected events.

To evaluate the system's macro-level performance, we conducted a cumulative statistical analysis of the 24-hour scheduling results. Using "Total Schedule Adjustment Time" as the core metric for measuring port operational stability, the results show a stark contrast. Constrained by linear thinking and cognitive limits, the manual schedule accumulated a total adjustment time of 31.0 hours, often leading to suboptimal congestion and queuing during high-pressure periods. Conversely, the globally optimized schedule generated by the system drastically compressed this metric to 9.5 hours. Overall calculations indicate that the agent achieved a 69.4% improvement in operational efficiency.

V. CONCLUSION

Addressing the highly dynamic and strongly constrained port channel scheduling problem, this paper proposes and validates an intelligent agent framework integrating semantic perception and LLMs.

Research results indicate that at the perception level, this framework achieves automatic mapping from raw AIS kinematic data to high-level navigation semantics via semantic middleware. It can autonomously identify speed anomalies and planned time drifts caused by external perturbations, effectively triggering the reconstruction mechanism.

At the decision level, the introduction of the LLM overcomes the limitations of traditional algorithms in handling unstructured domain rules (such as vessel task priorities and day-night window switching). Through cognitive reasoning, it realizes nonlinear global optimization based on vessel social value semantics, avoiding the "local lagging avoidance" common in manual scheduling. At the execution level, experimental results confirm that the agent can complete conflict-free scheduling within 40 seconds under high-stress navigation scenarios. Compared with the manual schemes of senior dispatchers, this framework reduces the Total Schedule Adjustment Time from 31.0 hours to 9.5 hours, achieving an optimization magnitude of 69.4%, thereby significantly enhancing system robustness and navigational efficiency.

Future research will focus on the deep integration of multi-source heterogeneous perception, planning to introduce multimodal data such as radar echoes and infrared vision into the perception chain to improve parsing accuracy under extreme weather conditions. Furthermore, we aim to explore collaborative gaming among multi-agents, constructing a large-scale distributed scheduling network covering the entire port area to address higher-dimensional shipping challenges.

REFERENCES

- [1] L. Agussurja, A. Kumar, and H. C. Lau, "Resource-constrained scheduling for maritime traffic management," in *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI-18)*, 2018, pp. 6086–6093.
- [2] L. Gan, Y. Wang, K. Li, and S. Liang, "Efficient online one-way traffic scheduling for restricted waterways," *Ocean Engineering*, vol. 237, p. 109515, 2021.
- [3] S. Q. Li and S. Jia, "The seaport traffic scheduling problem: Formulations and a column-row generation algorithm," *Transportation Research Part B: Methodological*, vol. 128, pp. 158–184, 2019.
- [4] J. Wang, K. Liu, Y. Yu, W. Wang, and X. Wu, "Vessel scheduling in multi-basin coastal ports affected by tidal currents," *Ocean Engineering*, vol. 325, p. 120822, 2025.
- [5] J. Aminu, R. Latip, Z. M. Hana, S. Kamarudin, and D. Gabi, "Systematic review of metaheuristic-based task scheduling strategies in edge computing environments," *Discover Computing*, vol. 28, p. 307, 2025.
- [6] M. Al Samrout, A. Sbihi, and A. Yassine, "An improved genetic algorithm for the berth scheduling with ship-to-ship transshipment operations integrated model," *Computers & Operations Research*, vol. 161, p. 106409, 2024.
- [7] Z. C. Cao, C. R. Lin, and M. C. Zhou, "A knowledge-based cuckoo search algorithm to schedule a flexible job shop with sequencing flexibility," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 1, pp. 56–69, 2021.
- [8] D. Pei, J. He, K. Liu, M. Chen, and S. Zhang, "Application of large language models and assessment of their ship-handling theory knowledge and skills for connected maritime autonomous surface ships," *Mathematics*, vol. 12, no. 15, p. 2381, 2024.

- [9] N. Chen, A. Yang, H. Wu, L. Chen, W. Xiong, and N. Jing, "SEMINT: An LLM-Empowered Long-Term Vessel Trajectory Prediction Framework," *International Journal of Geographical Information Science*, vol. 39, no. 9, pp. 1938–1972, 2025.
- [10] X. Guo, S. Yu, J. Zhang, H. Bi, X. Chen, and J. Liu, "A natural language-based automatic identification system trajectory query approach using large language models," *ISPRS International Journal of Geo-Information*, vol. 14, no. 5, p. 204, 2025.
- [11] S. Sun, L. Zhao, M. Deng, and X. Fu, "VTS-LLM: Domain-Adaptive LLM Agent for Enhancing Awareness in Vessel Traffic Services through Natural Language," 2025.
- [12] J. Kephart and D. Chess, "The vision of autonomic computing," *Computer*, vol. 36, no. 1, pp. 41–50, 2003.
- [13] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing Reasoning and Acting in Language Models," in *International Conference on Learning Representations (ICLR)*, 2023, arXiv:2210.03629.
- [14] C. A. Coello Coello, "Theoretical and numerical constraint-handling techniques used with evolutionary algorithms: A survey of the state of the art," *Computer Methods in Applied Mechanics and Engineering*, vol. 191, no. 11–12, pp. 1245–1287, Jan. 2002.
- [15] K. Wolsing, L. Roepert, J. Bauer, and K. Wehrle, "Anomaly detection in maritime AIS tracks: A review of recent approaches," *Journal of Marine Science and Engineering*, vol. 10, no. 1, p. 112, 2022.
- [16] K. Park, S. Sim, and H. Bae, "Vessel estimated time of arrival prediction system based on a path-finding algorithm," *Maritime Transport Research*, vol. 2, p. 100012, 2021.
- [17] A. Gatt and E. Krahmer, "Survey of the state of the art in natural language generation: Core tasks, applications and evaluation," *Journal of Artificial Intelligence Research*, vol. 61, pp. 65–170, 2018.
- [18] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474, arXiv:2005.11401.
- [19] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 24 824–24 837, arXiv:2201.11903.
- [20] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language Models Can Teach Themselves to Use Tools," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 68 539–68 551, arXiv:2302.04761.
- [21] G. Wang and C. Xu, "Simulation study on the capacity of tonggu channel in shenzhen port based on Flexsim," *Port & Waterway Engineering*, no. 9, pp. 181–185, Sep. 2019, serial No. 560; in Chinese.
- [22] X. Zhang, W. Guo, Z. Yang, J. Wang, and C. Wang, "Optimization of integrated accurate ride-tide planning and vessel scheduling in multi-functional ports with long channels," *Expert Systems with Applications*, vol. 273, p. 126894, May 2025.